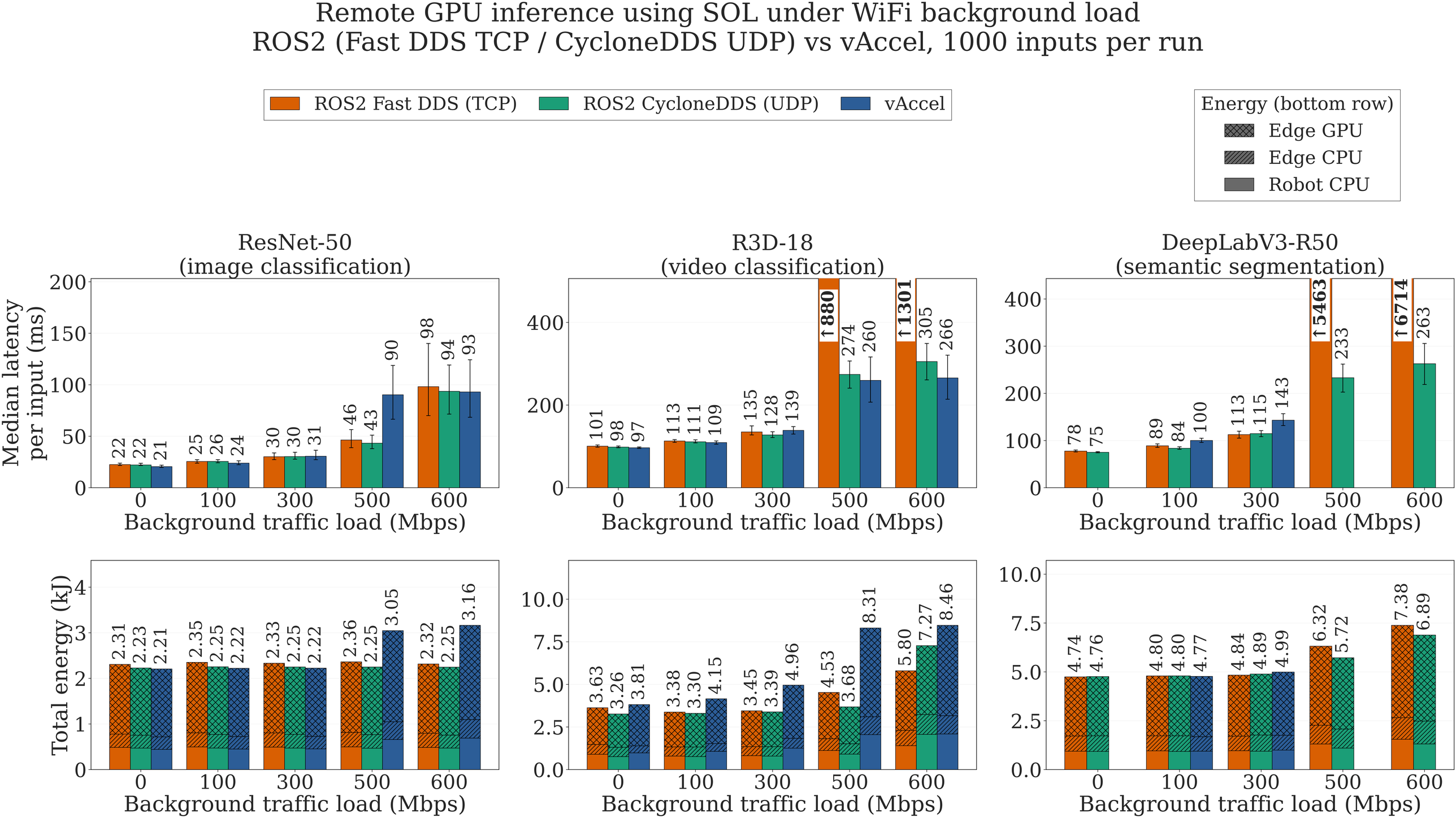
Remote GPU inference using SOL under WiFi background load
ROS2 (Fast DDS TCP / CycloneDDS UDP) vs vAccel, 1000 inputs per run
ROS2 Fast DDS (TCP)
ROS2 CycloneDDS (UDP)
vAccel
Energy (bottom row)
Edge GPU
Edge CPU
Robot CPU
ResNet-50
(image classification)
R3D-18
(video classification)
DeepLabV3-R50
(semantic segmentation)
Median latency per input (ms)
Total energy (kJ)
Background traffic load (Mbps)
0
100
300
500
600
22 22 21
25 26 24
30 30 31
46 43 90
98 94 93
101 98 97
113 111 109
135 128 139
↑880 274 260
↑1301 305 266
78 75
89 84 100
113 115 143
↑5463 233
↑6714 263
2.31 2.23 2.21
2.35 2.25 2.22
2.33 2.25 2.22
2.36 2.25 3.05
2.32 2.25 3.16
3.63 3.26 3.81
3.38 3.30 4.15
3.45 3.39 4.96
4.53 3.68 8.31
5.80 7.27 8.46
4.74 4.76
4.80 4.80 4.77
4.84 4.89 4.99
6.32 5.72
7.38 6.89